SUMMER AI WORKSHOP

AI for Coding / No-Code / Light Automation

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AI lets non-programmers become software engineers in one afternoon

meh.

AI lets beginners build small, bounded, useful prototypes if they can clearly describe the problem, inspect the result, test it, and iterate

GOOD TARGETS

Calculators:
GPA calculator
loan payment estimator
grade-weight calculator

GOOD TARGETS

Quizzes:
vocabulary quiz
study self-test
practice exam

GOOD TARGETS

Forms:
RSVP form
intake form
survey

GOOD TARGETS

Trackers:
habit tracker
assignment tracker
budget tracker

GOOD TARGETS

Simple webpages:
class resource page
Portfolio
event page

GOOD TARGETS

Simple webpages:
gradebook helper
inventory list
attendance summary

GOOD TARGETS

Tiny automations:
form response to email draft
spreadsheet cleanup
reminder list

BAD TARGETS

Login systems

Security and privacy complexity

BAD TARGETS

Payment apps

Real money, compliance, fraud risk

BAD TARGETS

Apps handling grades, medical data, or sensitive info

High consequence if wrong

BAD TARGETS

Multi-user real-time apps

State, syncing, edge cases

BAD **TARGETS**

"Build me Uber/TikTok/Amazon" ideas

Too many hidden systems

If the app can be explained with **3 to 5 inputs, 1 main action**, and a **visible output**, it is probably a good beginner AI build

AI as Pair Builder

NOT Magic Paste Machine

AI ROLES

Planner

Breaks the app into parts

AI ROLES

Builder

Writes first-draft code or no-code setup steps

AI ROLES

Explainer

Translates code/errors into plain language

AI ROLES

Tester

Suggests test cases

AI ROLES

Debugger

Helps diagnose failures

AI ROLES

Reviewer

Finds missing edge cases or confusing UI

HUMAN ROLES

Define the goal

AI cannot know what matters unless told

HUMAN ROLES

Check whether the output is useful

Plausible is not the same as correct

HUMAN ROLES

Run the app

Code that looks right may fail

HUMAN ROLES

Report exact errors

“It doesn’t work” is too vague

HUMAN ROLES

Decide tradeoffs

Simpler, safer, prettier, more flexible, faster

HUMAN ROLES

Stop scope creep

AI loves adding complexity

Spec -> AI plan

-> first version -> run it -> test it

-> report failures -> revise

-> retest -> document

BAD WORKFLOW

Ask vague prompt -> paste code -> hope

-> panic when broken

TEST

Good case

Normal valid input

TEST

Bad case

Wrong or *invalid* input

TEST

Empty case

Missing input

NO CODE

Best for Apps with forms, lists, simple workflows, dashboards

Fast, visual, less syntax

Platform limits, accounts, costs, less control

SINGLE-FILE WEBPAGE

Best for Calculators, quizzes, study tools, simple interactive pages

Teaches real app logic, easy to share as a file, flexible

No database, no login, more technical

FORM/SHEET AUTOMATION

Best for Surveys, signups, grading helpers, reports, repeated admin tasks

Familiar, data stored neatly, formulas help

Can become messy, privacy issues, fragile automations

DECISION

If the task is mostly... - > Choose...

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If the task is mostly... - > Choose...

Collecting responses -> Form plus sheet

DECISION

If the task is mostly... - > Choose...

Calculating or interacting on screen -> Single-file webpage

DECISION

If the task is mostly... - > Choose...

Tracking records with views -> No-code database/app

DECISION

If the task is mostly... - > Choose...

Repeating a low-risk admin step -> Light automation

DECISION

If the task is mostly... - > Choose...

Handling sensitive/private/high-stakes data -> Do not automate casually

LIGHT AUTOMATION

When this happens -> do this small action -> record or notify

When a form is submitted -> add row to sheet -> send confirmation email draft
When assignment dates are entered -> generate study reminders
When a CSV is uploaded -> clean columns -> produce summary table
When a student enters quiz answers -> score them -> show feedback

LIGHT AUTOMATION

Automate drafts, reminders, summaries, and copies **before** you automate **irreversible actions**